

Part X - Rheumatology and Immunology

Chapter 10E: Crystal Arthritis, Osteoarthritis, Fibromyalgia, and Regional Pain Syndromes

Medvale Internal Medicine - original synthesis for clinical learning and Graph-RAG extraction

This chapter covers the rheumatologic disorders that often behave differently from systemic autoimmune disease. Crystal arthritis can mimic septic arthritis and must be handled as an acute inflammatory arthritis. Osteoarthritis is common, mechanical, and structurally degenerative rather than primarily autoimmune. Fibromyalgia is a centralized pain syndrome with normal inflammatory testing. Regional pain syndromes localize to bursae, tendons, entheses, nerves, or periarticular structures rather than the joint space itself. The unifying clinical skill is localization: decide whether the pain is intra-articular, periarticular, inflammatory, mechanical, infectious, crystal-driven, neurologic, or centralized before ordering broad autoimmune panels.

High-yield orientation: do not let a positive imaging report or mildly abnormal laboratory value replace bedside localization. A painful swollen knee with fever is an aspiration problem before it is a serology problem. Chronic thumb-base pain with bony enlargement is usually osteoarthritis even if the ANA is weakly positive. Diffuse pain with fatigue and normal objective inflammation is usually not occult lupus. A patient with focal lateral shoulder pain may have rotator cuff disease rather than glenohumeral arthritis.

1. Core Localization: Joint, Periarticular Tissue, Crystal, Infection, or Central Pain

The first diagnostic branch is anatomic. True intra-articular disease produces pain with both active and passive range of motion, effusion, warmth, and limitation in multiple planes. Periarticular pain is usually worse with active movement or resisted testing but relatively preserved with passive movement. Tendinopathy, bursitis, ligament injury, enthesitis, nerve entrapment, and myofascial pain syndromes may be described by patients as joint pain, but the pathology sits outside the synovial space. This distinction determines whether joint aspiration, radiographs, ultrasound, physical therapy, injection, inflammatory markers, or neuropathic testing is the correct next step.

The second diagnostic branch is tempo. Hours to days suggests septic arthritis, crystal arthritis, trauma, hemarthrosis, or an acute inflammatory flare. Weeks to months suggests osteoarthritis progression, inflammatory arthritis, tendon overuse, or early systemic disease. Years of waxing and waning pain with normal objective inflammation suggests mechanical or centralized pain. The third branch is distribution: a single hot joint is an emergency until infection is excluded; symmetric small-joint inflammatory arthritis suggests RA; asymmetric lower-extremity arthritis suggests spondyloarthritis or crystal disease; diffuse pain above and below the waist with sleep disturbance suggests fibromyalgia.

Clinical problem	Immediate questions	Most useful next step	Common traps
Single hot swollen joint	Fever? immunosuppression? prosthetic joint? overlying cellulitis? anticoagulation? prior gout?	Arthrocentesis for cell count/differential, Gram stain/culture, and crystals before anchoring on gout.	Serum uric acid does not diagnose an acute flare; crystals do not rule out infection.
Multiple painful joints	Inflammatory stiffness >30-60 min? swelling? rash? mucosal ulcers? psoriasis? IBD? uveitis?	Focused exam plus ESR/CRP, CBC/CMP/urinalysis, disease-specific serology only when pretest probability is real.	Ordering ANA/RF/ANCA as screening tests for nonspecific pain generates false positives.
Mechanical joint pain	Worse with use? improves with rest? brief stiffness? crepitus? bony enlargement?	Weight-bearing radiographs when diagnosis or severity is uncertain; treat function and pain.	MRI often shows age-related abnormalities that do not identify the pain generator.
Periarticular pain	Pain with resisted motion? focal tenderness over bursa/tendon? passive ROM preserved?	Targeted maneuvers, ultrasound if uncertain, activity modification/PT, selective injection.	Calling every painful shoulder or hip arthritis misses bursitis, tendinopathy, and radiculopathy.
Diffuse chronic pain	Sleep? fatigue? cognitive symptoms? mood? exercise intolerance? objective synovitis?	Diagnose clinically after excluding inflammatory, endocrine, neurologic, and medication causes suggested by history/exam.	Repeated autoimmune panels reinforce illness anxiety and rarely clarify fibromyalgia.

2. Crystal Arthritis: General Principles

Crystal arthritis is inflammation triggered by deposition of crystals in or around joints. The two classic synovial-fluid diagnoses are monosodium urate crystals in gout and calcium pyrophosphate crystals in CPPD. Both can produce abrupt severe pain, erythema, warmth, swelling, and marked functional limitation. Both can raise ESR and CRP. Both can coexist with chronic degenerative joint disease. Both can be mistaken for cellulitis or septic arthritis. The practical lesson is that acute crystal arthritis is a clinical mimic of infection, and aspiration remains the highest-yield diagnostic test when the presentation is atypical, severe, febrile, immunosuppressed, prosthetic, or first-ever.

Synovial fluid microscopy is diagnostic when crystals are seen, but absence of crystals in a small or poorly handled sample does not completely exclude crystal disease. Polarized microscopy distinguishes gout from CPPD by crystal shape and birefringence. Culture is still required when infection is possible because septic arthritis can coexist with crystal arthritis. Serum markers are supportive rather than definitive: leukocytosis, ESR, and CRP rise in infection and crystal flares; serum urate may be normal during an acute gout attack; radiographic chondrocalcinosis supports CPPD but is not required for CPPD and is not always symptomatic.

Feature	Gout	CPPD / pseudogout	Septic arthritis comparison
Crystal	Monosodium urate; needle-shaped; negatively birefringent under polarized microscopy.	Calcium pyrophosphate; rhomboid or rod-shaped; weakly positively birefringent.	No diagnostic crystal; Gram stain may be negative, so culture is essential.
Typical patient pattern	Middle-aged or older adults; risk increased by CKD, diuretics, obesity/metabolic syndrome, alcohol, high purine intake, cyclosporine/tacrolimus, and genetic predisposition.	Older adults; associated with osteoarthritis, prior joint injury, hemochromatosis, hyperparathyroidism, hypomagnesemia, and sometimes hypothyroidism.	Any age with bacteremia, skin infection, injection drug use, prosthetic joint, immunosuppression, advanced age, or preexisting joint disease.
Common joints	First MTP podagra, midfoot, ankle, knee, wrist, olecranon bursa; chronic tophaceous disease may involve many sites.	Knee, wrist, shoulder, ankle, elbow; can mimic OA in unusual joints such as wrist or MCP.	Knee most common native joint; hip, shoulder, sternoclavicular, sacroiliac, or prosthetic joint can occur.
Radiography	Early films may be normal. Chronic gout can show punched-out erosions with overhanging edges and preserved joint space until late.	Chondrocalcinosis, especially menisci, triangular fibrocartilage, cartilage, or symphysis pubis; OA-like changes may occur in atypical distribution.	May be normal early; late joint-space loss/destruction. Imaging must not delay aspiration if joint is accessible.
Acute therapy	NSAID, colchicine, or glucocorticoid depending on comorbidities and timing; ice/rest. Continue established ULT during flare.	NSAID, colchicine, intra-articular or systemic glucocorticoid; treat associated metabolic contributors.	Prompt antibiotics after cultures/aspiration and urgent drainage when indicated.
Chronic prevention	Treat-to-target urate-lowering therapy when indicated; goal serum urate generally <6 mg/dL, lower in severe tophaceous disease.	No proven crystal-removing therapy; recurrence prevention with low-dose colchicine in selected patients; manage OA phenotype.	Not a chronic crystal disease; evaluate source and complications.

3. Gout

Pathophysiology and risk factors

Gout develops when monosodium urate crystals form in tissues because body urate stores exceed solubility. Hyperuricemia is necessary for most gout but is not sufficient: many hyperuricemic patients never have gout, and acute gout can occur with a normal serum urate measurement. The final inflammatory event is innate immune activation around crystals, especially neutrophil-rich synovitis. Flares may be triggered by dehydration, illness, surgery, trauma, heavy alcohol intake, abrupt dietary changes, or abrupt changes in urate concentration after starting urate-lowering therapy.

Most gout reflects underexcretion of urate rather than overproduction. Chronic kidney disease and drugs that decrease renal urate excretion are major drivers. Thiazide and loop diuretics are common medication contributors. Low-dose aspirin may raise urate but is usually continued when cardiovascular indications are strong. Calcineurin inhibitors can increase risk. High-fructose beverages, beer, spirits, red meat, and shellfish can contribute, while weight loss and metabolic risk control reduce flare burden. Dietary advice should be framed as adjunctive rather than as a substitute for urate-lowering therapy in recurrent or tophaceous gout.

Clinical phenotypes

Classic podagra is abrupt, exquisitely painful inflammation of the first metatarsophalangeal joint. Patients may be unable to tolerate a sheet touching the foot. Flares often peak within 24 hours and gradually resolve over days to two weeks if untreated. Recurrent gout may involve ankles, knees, wrists, fingers, elbows, bursae, and tendons. Chronic tophaceous gout produces firm urate deposits in the helix of the ear, olecranon bursa, fingers, toes, Achilles tendon, or other soft tissues. Tophi may ulcerate, drain chalky material, erode bone, compress nerves, or mimic nodules from other rheumatic diseases.

Intercritical gout refers to asymptomatic periods between flares. The absence of pain during intercritical periods does not mean the disease has resolved; crystal burden can persist and progressively damage joints. Chronic gouty arthropathy may look like destructive inflammatory arthritis or erosive osteoarthritis. In older adults, women after menopause, patients with CKD, and patients taking diuretics, gout may present as polyarticular disease rather than classic podagra.

Diagnosis

The most definitive diagnosis is identification of monosodium urate crystals in synovial fluid or tophus aspirate. A clinical diagnosis can be reasonable in recurrent classic podagra with hyperuricemia, but first presentations, atypical joints, fever, immunosuppression, prosthetic joints, or diagnostic uncertainty should prompt aspiration. Serum urate supports long-term management but is not a rule-in or rule-out test during acute flares. A very high urate level raises probability, but a normal value during inflammation does not exclude gout because urate can fall during acute illness.

Imaging is most useful when aspiration is not possible or chronic disease is suspected. Ultrasound may show a double-contour sign from urate on cartilage or tophaceous deposits. Dual-energy CT can identify urate deposition, especially in difficult-to-aspirate locations, but false positives and availability limitations remain. Plain radiographs are often normal early and become helpful later when punched-out erosions, overhanging edges, and tophi are present.

Red flag: never diagnose septic arthritis as just gout because the patient has prior gout. Aspirate if the flare is unusually severe, associated with high fever, bacteremia risk, immunosuppression, prosthetic material, overlying skin infection, or failure to improve as expected.

Acute gout treatment

Acute treatment should start early, ideally within the first 24 hours. NSAIDs, colchicine, and glucocorticoids are all appropriate first-line options; the best choice depends on renal function, GI bleeding risk, anticoagulation, heart failure, diabetes, infection concern, drug interactions, and number of involved joints. Intra-articular glucocorticoid injection is useful for one or two accessible joints after infection is excluded. Systemic glucocorticoids are often used when NSAIDs and colchicine are unsafe, such as in advanced CKD or anticoagulated patients, but they can worsen hyperglycemia, fluid retention, mood symptoms, and infection risk.

Colchicine is most effective early and is limited by diarrhea, nausea, renal impairment, hepatic impairment, and CYP3A4 or P-glycoprotein drug interactions. Toxicity can be severe with clarithromycin, certain azoles, cyclosporine, some calcium channel blockers, and renal dysfunction. NSAIDs are effective but should be avoided or used cautiously in CKD, peptic ulcer disease, heart failure, uncontrolled hypertension, anticoagulation, and high cardiovascular risk. Do not start antibiotics for a hot joint without considering aspiration and cultures; do not inject steroids into a potentially infected joint.

Urate-lowering therapy and prophylaxis

Urate-lowering therapy is used to reduce total body urate stores and prevent future flares, tophi, and joint damage. Indications include tophaceous gout, radiographic gout damage, frequent flares, and selected high-risk first presentations such as very high serum urate, CKD, or urolithiasis. Modern gout management uses a treat-to-target strategy rather than fixed-dose therapy: titrate urate-lowering medication to achieve and maintain serum urate below target, commonly <6 mg/dL. Allopurinol is first-line for most patients, including many with CKD, but it should be started low and titrated while monitoring renal function, liver tests, CBC when appropriate, rash, and hypersensitivity risk. Febuxostat is another xanthine oxidase inhibitor but requires cardiovascular risk consideration. Probenecid is less useful in CKD and is avoided in uric acid nephrolithiasis.

Starting urate-lowering therapy can precipitate flares because changing urate levels mobilizes crystals. Anti-inflammatory prophylaxis with low-dose colchicine, low-dose NSAID, or low-dose prednisone in selected patients is commonly used during initiation and titration. Established urate-lowering therapy should generally be continued during an acute flare; stopping and restarting creates urate fluctuation and can worsen the cycle. Patient education matters because early flares after starting therapy are not treatment failure; they are a predictable mobilization phenomenon that decreases as crystal burden falls.

Decision point	Practical approach	Graph-RAG association hooks
When to aspirate	First flare, atypical flare, fever, severe leukocytosis, immunosuppression, prosthetic joint, concern for septic arthritis, or uncertain diagnosis.	Hot joint -> arthrocentesis -> crystals and culture.
When to start ULT	Tophi, radiographic damage, frequent flares, or selected high-risk patients. Shared decision-making for less frequent disease.	Recurrent gout -> urate burden -> ULT.
Allopurinol start	Start low, titrate to urate target, monitor for rash/hypersensitivity and labs. Consider HLA-B*58:01 testing in higher-risk ancestry groups per local practice/guidelines.	Xanthine oxidase inhibition -> lower urate production.
Prophylaxis	Use colchicine/NSAID/low-dose steroid when starting ULT if safe; continue long enough to cover mobilization flares.	ULT initiation -> flare risk -> prophylaxis.
Lifestyle	Weight loss if indicated, limit beer/spirits/high-fructose drinks, reduce high-purine excess, review diuretics, manage CKD/metabolic syndrome.	Metabolic syndrome/CKD/diuretics -> hyperuricemia -> gout.

4. Calcium Pyrophosphate Deposition Disease (CPPD, Pseudogout)

CPPD is caused by calcium pyrophosphate crystal deposition in cartilage and periarticular tissues. It is strongly age-associated and may be clinically silent, present as acute pseudogout, mimic osteoarthritis, or rarely resemble chronic inflammatory arthritis. The knee and wrist are classic sites. A first attack in a younger patient should prompt evaluation for associated metabolic disorders, especially hemochromatosis, hyperparathyroidism, hypomagnesemia, and sometimes hypothyroidism. Joint trauma and surgery can trigger attacks.

Acute CPP crystal arthritis resembles gout and septic arthritis: abrupt pain, swelling, warmth, and reduced range of motion. Unlike gout, first MTP podagra is less classic; knee and wrist disease are particularly common. Radiographs may show chondrocalcinosis, but radiographic chondrocalcinosis can be asymptomatic and its absence does not exclude CPPD. Synovial fluid analysis showing weakly positively birefringent rhomboid crystals is the diagnostic anchor.

Treatment is symptomatic because no established therapy dissolves CPP crystals or reliably prevents deposition. Acute therapy includes joint aspiration, intra-articular steroid if infection is excluded, NSAIDs if safe, colchicine, or systemic glucocorticoids when multiple joints or contraindications exist. Chronic recurrent CPPD may be managed with low-dose colchicine or anti-inflammatory strategies, plus treatment of associated metabolic abnormalities. CPPD often overlaps with osteoarthritis; the pain generator may be crystal inflammation, structural OA, or both.

CPPD phenotype	Clinical picture	Key diagnostic clue	Management focus
Asymptomatic chondrocalcinosis	Radiographic cartilage calcification found incidentally.	No inflammatory attacks; often older adult.	No crystal-specific treatment; avoid overdiagnosis.
Acute CPP crystal arthritis	Sudden hot swollen knee/wrist/ankle/shoulder; can be febrile.	Synovial rhomboid weakly positive birefringent crystals; culture to exclude infection.	NSAID/colchicine/steroid chosen by comorbidity; aspirate if uncertain.
OA with CPPD	Chronic mechanical pain with OA changes, sometimes in atypical joints such as wrist, MCP, shoulder, ankle.	Chondrocalcinosis plus OA distribution not typical for primary OA.	Treat OA function/pain and manage inflammatory flares separately.
Chronic inflammatory CPPD	Persistent or recurrent inflammatory polyarthritis that can mimic RA.	Older age, chondrocalcinosis, crystals, usually lacks highly specific RA pattern/serology.	Colchicine or anti-inflammatory therapy; avoid unnecessary lifelong RA labeling.

5. Other Crystal and Peri-crystal Syndromes

Basic calcium phosphate deposition can cause calcific periarthritis and calcific tendinitis, especially around the shoulder. Hydroxyapatite crystals are too small for routine compensated polarized microscopy, so diagnosis is often clinical and radiographic. Acute calcific tendinitis can cause intense focal pain and limited motion, commonly in the rotator cuff. Milwaukee shoulder is a destructive arthropathy classically associated with basic calcium phosphate crystals, large effusions, rotator cuff degeneration, and older age. Treatment is supportive and local: analgesia, physical therapy, aspiration when large effusion is present, and selected injection or orthopedic referral.

Cholesterol crystals may appear in chronic inflammatory effusions but are rarely the central board-style diagnosis. Oxalate crystal arthropathy can occur in advanced kidney disease or primary oxalosis. The practical high-yield point is not to memorize rare crystals at the expense of the emergency: an acutely inflamed joint must be evaluated for infection, gout, CPPD, trauma, and hemarthrosis before rare etiologies are considered.

6. Osteoarthritis

Conceptual model

Osteoarthritis is not simply wear and tear. It is a whole-joint disorder involving cartilage, subchondral bone, synovium, menisci, ligaments, capsule, periarticular muscle, and pain-processing pathways. Mechanical stress, age, prior injury, malalignment, obesity, genetics, and local inflammation interact to produce cartilage matrix breakdown, osteophyte formation, subchondral sclerosis, cysts, episodic synovitis, and functional decline. OA pain is usually activity-related, worse with use, relieved by rest, and associated with brief morning stiffness or gelling after inactivity. Stiffness lasting hours, marked warmth, systemic symptoms, and high inflammatory markers should prompt reconsideration.

The severity of radiographic OA and the severity of pain do not always match. Some patients have severe x-ray changes with modest pain; others have severe pain with mild structural findings because synovitis, bone marrow lesions, meniscal pathology, central sensitization, mood, sleep, and muscle weakness contribute. The clinical goal is not to normalize imaging; it is to improve pain, function, mobility, sleep, strength, and participation while recognizing when surgery or another diagnosis is needed.

Clinical patterns

Hand OA commonly affects the distal interphalangeal joints, proximal interphalangeal joints, and first carpometacarpal joint. DIP bony enlargement is called Heberden nodes; PIP bony enlargement is called Bouchard nodes. First CMC OA causes thumb-base pain with pinch, grip, jars, keys, and writing. MCP and wrist predominance should raise suspicion for RA, CPPD, hemochromatosis arthropathy, or another inflammatory/metabolic process rather than ordinary primary hand OA.

Knee OA causes pain with stairs, walking, rising from a chair, squatting, or prolonged standing. Exam may show crepitus, bony enlargement, small effusion, varus or valgus deformity, quadriceps weakness, and reduced extension. Hip OA causes groin, buttock, lateral hip, or anterior thigh pain and reduced internal rotation. True hip OA may present as knee pain, so hip range of motion must be examined in patients with unexplained knee pain. Spine OA and degenerative disc/facet disease can produce axial pain, radiculopathy, or spinal stenosis, but these belong partly to neurology/orthopedics rather than inflammatory arthritis.

Diagnosis and imaging

OA is usually a clinical diagnosis supported by plain radiographs when diagnosis is uncertain, symptoms are severe, surgery is considered, trauma is possible, or another disease is in the differential. Weight-bearing knee films are more informative than non-weight-bearing films. Classic x-ray features include asymmetric joint-space narrowing, osteophytes, subchondral sclerosis, and subchondral cysts. MRI is not routinely needed for typical OA because it often detects incidental meniscal tears, cartilage defects, and degenerative changes that do not change management.

Laboratory tests are generally normal in OA. ESR and CRP are not needed for classic mechanical OA but are appropriate when symptoms suggest inflammatory arthritis, infection, polymyalgia rheumatica, malignancy, or systemic disease. Synovial fluid in OA is noninflammatory, usually with low WBC count and a predominance of mononuclear cells; however, OA can coexist with CPPD or gout, so crystals should be checked when effusion is acute or inflammatory.

Feature	Osteoarthritis	Inflammatory arthritis clue	Why it matters
Pain timing	Worse with use; improves with rest; brief stiffness usually <30 min.	Prolonged morning stiffness, nocturnal inflammatory pain, swelling/warmth.	Inflammatory timing changes diagnostic pathway and medication choice.
Joint distribution	DIP, PIP, first CMC, knees, hips, spine; asymmetric or weight-bearing predominance.	MCP/wrist symmetric swelling, MTP squeeze pain, sacroiliitis, enthesitis, dactylitis.	Distribution often outperforms screening serology.
Exam texture	Bony enlargement, crepitus, reduced ROM, mild effusion possible.	Boggy synovitis, warmth, tendon sheath swelling, nodules, rash.	Bony versus soft synovial swelling separates OA from RA/crystal/infection.
Labs	Usually normal ESR/CRP, negative disease-specific serologies.	High ESR/CRP, anemia, thrombocytosis, autoantibodies with compatible syndrome.	Positive ANA/RF alone does not diagnose inflammatory disease.
Imaging	Osteophytes, asymmetric joint-space loss, sclerosis, cysts.	Erosions, periarticular osteopenia, sacroiliitis, chondrocalcinosis, infection changes.	Different radiographic patterns encode different mechanisms.

Treatment principles

OA treatment should be multimodal and individualized. Exercise is foundational: aerobic activity, strengthening, neuromuscular training, range-of-motion work, and balance training improve pain and function even when imaging is unchanged. Weight loss improves knee and hip symptoms in patients with overweight or obesity; even modest loss can reduce joint load, and larger sustained loss often produces larger benefit. Bracing, splints, canes, footwear changes, occupational therapy, and physical therapy should be framed as disease management rather than failure of medication.

Topical NSAIDs are preferred for many patients with hand or knee OA because systemic exposure is lower than oral NSAIDs. Oral NSAIDs are effective but require risk assessment: kidney disease, hypertension, heart failure, coronary disease, anticoagulation, GI bleeding, older age, and interacting drugs change the risk-benefit balance. Acetaminophen has modest benefit and is limited by liver toxicity at high doses or with alcohol use. Duloxetine can help OA pain, especially when central sensitization, mood symptoms, or widespread pain coexist. Intra-articular glucocorticoid injection can provide short-term relief for knee or hip OA flares; hip injection should be image-guided. Opioids are generally avoided for chronic OA except rare circumstances; tramadol has limited conditional roles but still carries dependence, fall, serotonin, and cognitive risks.

Surgical referral is appropriate when pain and disability remain severe despite optimized conservative therapy, radiographs show advanced structural disease compatible with symptoms, and the patient is medically and psychologically prepared for postoperative rehabilitation. Total hip and knee arthroplasty can be transformative for advanced disease. Conversely, arthroscopic surgery for degenerative meniscal tears in the setting of OA is often low value unless there is true mechanical locking or another specific surgical indication.

Intervention category	Examples	Best fit / caution
Education and self-management	Explain OA as whole-joint disease; pacing, flare plan, sleep, realistic goals.	Reduces fear-avoidance and improves adherence.
Exercise/PT	Strengthening, aerobic exercise, aquatic exercise, balance, neuromuscular training.	Core therapy for knee/hip OA; start low and progress to avoid flare discouragement.
Weight and biomechanics	Weight loss, cane in contralateral hand, tibiofemoral brace, first CMC orthosis, footwear.	Especially useful in knee/hip/first CMC OA; mechanics can lower pain without changing x-ray.
Topical medication	Topical NSAID, capsaicin selected cases.	Good for hand/knee OA; fewer systemic NSAID risks.

Oral medication	NSAID if safe, acetaminophen limited, duloxetine for selected chronic pain phenotype.	Assess renal/CV/GI risks; avoid chronic steroid pills for OA.
Injection	Intra-articular glucocorticoid for short-term relief; image-guided hip injection.	Avoid injection through cellulitis; consider diabetes hyperglycemia and frequent-injection harms.
Surgery	Arthroplasty for advanced hip/knee OA with severe functional limitation.	Use when symptoms, imaging, goals, and medical readiness align.

7. Erosive Hand OA and OA Mimics

Erosive hand OA is an inflammatory phenotype of hand OA that can cause abrupt pain, swelling, erythema, and central erosions in interphalangeal joints. It may mimic psoriatic arthritis or RA, but the distribution often favors DIP and PIP joints with relative MCP sparing. Radiographs may show central erosions with a gull-wing or sawtooth pattern rather than marginal erosions typical of RA. Management remains mainly symptomatic: topical/oral NSAIDs if safe, hand therapy, splinting, and function preservation; disease-modifying immunosuppression has not shown the same role as in RA.

Important OA mimics include CPPD arthropathy affecting wrists or MCPs, hemochromatosis arthropathy affecting second and third MCPs with hook-like osteophytes, psoriatic arthritis affecting DIPs with nail disease/dactylitis, RA affecting MCP/PIP/wrists with prolonged stiffness and soft synovitis, and neuropathic arthropathy with disproportionate destruction and reduced protective sensation. These mimics are why distribution, exam texture, inflammatory timing, skin/nail findings, and radiograph pattern matter more than isolated labels.

8. Fibromyalgia and Centralized Pain

Conceptual model

Fibromyalgia is a chronic centralized pain syndrome characterized by widespread pain, fatigue, nonrestorative sleep, cognitive symptoms, and multiple somatic symptoms. It is not an inflammatory arthritis, not a structural joint disease, and not a diagnosis of malingering. The central abnormality is altered pain processing and amplification, often interacting with poor sleep, stress physiology, mood disorders, trauma history, autonomic symptoms, deconditioning, and comorbid painful disorders. Patients may describe pain in joints, muscles, tendons, and bones, but objective synovitis, weakness from muscle inflammation, destructive imaging changes, or high inflammatory markers are absent unless a second disease coexists.

Fibromyalgia commonly coexists with autoimmune disease, osteoarthritis, hypermobility, migraine, irritable bowel syndrome, interstitial cystitis/bladder pain syndrome, temporomandibular disorder, depression, anxiety, PTSD, and sleep apnea. Coexistence creates diagnostic pitfalls. A patient with lupus can also have fibromyalgia pain that does not respond to steroids. A patient with OA can have pain severity out of proportion to x-ray changes because central sensitization has become part of the pain state. The task is not to choose between real disease and fibromyalgia; it is to identify all contributors and avoid escalating immunosuppression for noninflammatory pain.

Diagnosis

Diagnosis is clinical and based on widespread pain plus associated symptom burden after evaluating for alternative explanations suggested by history and exam. Earlier tender-point counts are less central in modern practice than the overall pattern: generalized pain in multiple regions, fatigue, poor sleep, cognitive complaints, headaches, paresthesias without objective neurologic deficit, and normal inflammatory evaluation. Basic screening may include CBC, CMP, TSH, ESR/CRP, CK, vitamin D or B12 depending on context, and sleep/mood assessment; broad ANA/RF/ANCA testing is not indicated unless there are specific autoimmune features.

Red flags against uncomplicated fibromyalgia include objective joint swelling, persistent fever, unintentional weight loss, focal neurologic deficit, true proximal muscle weakness, elevated CK, hematuria/proteinuria, inflammatory back pain with uveitis/IBD/psoriasis, high ESR/CRP without explanation, new cancer concern, or progressive localized bone pain. Fibromyalgia can be diagnosed positively, not merely by exhaustion after every possible test, but red flags must be respected.

Domain	Fibromyalgia pattern	Concerning alternative
Pain	Widespread, migratory or diffuse, muscles and joints, allodynia, flares with stress/sleep loss.	Single destructive joint, focal bone pain, inflammatory swollen joints, radicular dermatomal pain.
Stiffness	Subjective stiffness without objective synovitis; often all day and linked to poor sleep.	Morning stiffness with boggy synovitis, warmth, effusion, elevated inflammatory markers.
Fatigue/sleep	Nonrestorative sleep, insomnia, possible sleep apnea/restless legs.	Anemia, renal/hepatic disease, hypothyroidism, inflammatory disease, malignancy.
Neurologic/cognitive	Brain fog, headaches, paresthesias with normal exam.	Focal weakness, myelopathy, seizure, objective neuropathy, CNS inflammatory disease.

Labs	Usually normal ESR/CRP/CK; incidental low-titer ANA may occur in general population.	High inflammatory markers, cytopenias, renal urine sediment, high CK, disease-specific serology with syndrome.
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Management

The most effective management begins with explanation. Patients need to hear that the pain is real, common, and potentially improvable, but it is driven by nervous-system amplification rather than joint destruction. A successful plan emphasizes graded activity, sleep restoration, stress modulation, treatment of mood and trauma-related conditions, and management of comorbid pain generators. Exercise should start below the flare threshold and progress slowly; abrupt intense exercise often worsens symptoms and reinforces avoidance. Cognitive behavioral therapy, mindfulness-based approaches, acceptance and commitment therapy, and pain neuroscience education can reduce disability and fear even when pain persists.

Medications can help selected patients but are adjuncts. Duloxetine, milnacipran, pregabalin, low-dose tricyclic antidepressants, or gabapentin-like strategies may improve pain, sleep, or mood depending on the phenotype and contraindications. NSAIDs and opioids are usually ineffective for core fibromyalgia pain, though NSAIDs may help a coexisting inflammatory or mechanical generator. Chronic opioids are particularly problematic because they worsen hyperalgesia, sleep-disordered breathing, constipation, cognition, falls, and dependence risk. Steroids and immunosuppressants should not be used for fibromyalgia itself.

9. Regional Pain Syndromes: A Practical Map

Regional pain syndromes are focal musculoskeletal pain generators outside the joint space. They are common, clinically testable, and often more responsive to physical therapy, ergonomic correction, local injection, splinting, or nerve-directed management than to systemic rheumatologic therapy. The key is to reproduce the patient's pain with a structure-specific maneuver. Passive joint range is often preserved compared with true arthritis, although severe pain can create guarding.

Region / syndrome	Typical clues	Exam localization	Initial management
Rotator cuff tendinopathy / impingement	Lateral shoulder pain, overhead activity pain, night pain when lying on side.	Painful arc, weakness/pain with empty-can or external rotation; passive ROM relatively preserved unless frozen shoulder.	Activity modification, PT emphasizing scapular/rotator cuff strength, NSAID/topical, subacromial injection selected cases.
Adhesive capsulitis	Gradual shoulder stiffness and pain; diabetes/thyroid disease risk.	Both active and passive ROM restricted, especially external rotation.	Stretching/PT, intra-articular steroid early, time course months-years; evaluate diabetes.
Olecranon bursitis	Posterior elbow swelling; may be painless, traumatic, gouty, RA-related, or septic.	Fluctuant swelling over olecranon; elbow joint ROM often preserved.	Compression/protection if aseptic; aspirate if infection/crystal concern; antibiotics if septic.
Lateral epicondylitis	Lateral elbow pain with gripping, lifting, racket/tool use.	Pain with resisted wrist extension or middle-finger extension.	Load modification, counterforce brace, eccentric rehab; avoid repeated steroid injections.
De Quervain tenosynovitis	Radial wrist/thumb pain, postpartum or repetitive thumb use.	Pain over first dorsal compartment; Finkelstein/Eichhoff maneuver reproduces pain.	Thumb spica splint, NSAID/topical, corticosteroid injection if persistent.
Carpal tunnel syndrome	Nocturnal paresthesias thumb-index-middle; shaking hand helps.	Median distribution sensory symptoms, thenar weakness late; Phalen/Tinel may help.	Night splint, ergonomic changes, steroid injection selected, EMG/surgery if severe/denervation.
Greater trochanteric pain syndrome	Lateral hip pain, worse lying on side/stairs; often gluteal tendinopathy.	Tender greater trochanter, pain with resisted hip abduction; hip joint ROM usually less groin-limited.	Hip abductor strengthening, gait mechanics, weight/load modification, injection selected.
Pes anserine bursitis/tendinopathy	Medial proximal tibial pain in knee OA, obesity, runners.	Tenderness below medial joint line, not exactly intra-articular line.	PT, hamstring/adductor flexibility, OA management, topical/oral NSAID if safe.
Plantar fasciitis	First-step heel pain, improves then worsens with prolonged standing.	Tender medial calcaneal tubercle; pain with dorsiflexion stretch.	Stretch plantar fascia/calf, night splint, orthotics, load reduction.

Bursitis: aseptic, crystal, inflammatory, or septic

Bursae reduce friction between tendons, skin, and bone. Bursitis can result from repetitive pressure, trauma, gout, CPPD, RA, infection, or systemic inflammation. Olecranon and prepatellar bursae are superficial and prone to trauma and infection. Septic bursitis should be suspected when there is warmth, erythema, tenderness, fever, skin break, immunosuppression, or occupational/recreational kneeling or elbow pressure. Unlike septic arthritis, the adjacent joint may retain relatively preserved passive range of motion. Aspiration of a superficial bursa is considered when infection or crystals are possible; steroid injection should be avoided if infection has not been excluded.

Tendinopathy and enthesopathy

Tendinopathy is failed tendon adaptation to load and microinjury; it is often degenerative rather than inflammatory. Pain is reproduced by loading the tendon, not necessarily by passive joint motion. Enthesopathy refers to pathology at tendon or ligament insertion into bone. Mechanical enthesopathy occurs from repetitive stress, while inflammatory enthesitis is a hallmark of spondyloarthritis and is associated with morning stiffness, improvement with movement, psoriasis, IBD, uveitis, dactylitis, or sacroiliitis. Treatment differs: mechanical tendinopathy needs load management and progressive strengthening, while inflammatory enthesitis may require systemic rheumatologic therapy.

Nerve entrapment masquerading as joint pain

Nerve entrapments can be perceived as deep joint or limb pain. Carpal tunnel syndrome, ulnar neuropathy at the elbow, meralgia paresthetica, tarsal tunnel syndrome, cervical radiculopathy, and lumbar radiculopathy should be considered when pain is burning, electric, associated with numbness/tingling, follows a nerve distribution, worsens with position, or has weakness/reflex changes. Rheumatologic disease can contribute through synovitis, tenosynovitis, deformity, or edema compressing nerves, but the treatment target becomes decompression, splinting, nerve studies, or surgery rather than simply increasing anti-inflammatory therapy.

10. Integrated Diagnostic Algorithms

Algorithm A: acute monoarthritis

Step 1: decide whether this is a true joint effusion or periarticular swelling. If passive range of motion is painful and restricted in multiple planes, treat it as intra-articular. Step 2: check for emergency features: fever, bacteremia risk, prosthetic joint, immunosuppression, severe pain with inability to bear weight, overlying cellulitis, recent procedure, or very high inflammatory markers. Step 3: aspirate when septic arthritis is plausible. Send synovial WBC/differential, Gram stain, culture, and crystal analysis. Blood cultures are useful when febrile or bacteremic. Step 4: begin antibiotics after cultures when septic arthritis is suspected; do not wait for crystal results if the patient is toxic. Step 5: if crystals are found and cultures are negative with low infection concern, treat gout or CPPD based on comorbidities. Step 6: if bloody fluid, consider trauma, anticoagulation, hemophilia, tumor, or vascular injury.

Algorithm B: chronic knee pain

Chronic knee pain should be separated into intra-articular mechanical pain, inflammatory arthritis, referred pain, and periarticular pain. OA is likely when pain is worse with stairs or walking, stiffness is brief, crepitus or bony enlargement is present, and risk factors such as age, obesity, prior injury, or malalignment exist. Weight-bearing radiographs help stage disease. Inflammatory arthritis is more likely with prolonged morning stiffness, warmth, recurrent large effusions, other swollen joints, psoriasis, IBD, uveitis, or high ESR/CRP. Referred pain from hip OA should be considered when knee x-ray is mild but hip internal rotation is painful or restricted. Pes anserine pain, patellar tendinopathy, iliotibial band pain, and radiculopathy are common non-articular mimics.

Algorithm C: widespread pain

For widespread pain, first look for objective inflammatory, endocrine, neurologic, malignant, medication-related, or infectious disease. History and exam should ask about synovitis, rash, Raynaud, oral ulcers, proximal weakness, neuropathy, renal findings, fevers, weight loss, sleep, mood, trauma, substance use, and medication exposures such as statins or aromatase inhibitors. If objective inflammation is absent and the pattern is chronic widespread pain with fatigue, nonrestorative sleep, cognitive symptoms, headaches, IBS-like symptoms, and normal basic evaluation, diagnose fibromyalgia positively and begin a rehabilitation-centered plan. If there is one focal severe pain generator plus generalized sensitivity, treat both rather than attributing all pain to one label.

11. High-Yield Tables for Exam and Clinical Reasoning

Finding	Most likely association	Pearl
Needle-shaped negatively birefringent crystals	Gout	Serum urate may be normal during flare.
Rhomboid weakly positively birefringent crystals	CPPD	Evaluate metabolic causes if young or atypical.
Chondrocalcinosis of meniscus or triangular fibrocartilage	CPPD	Can be asymptomatic; correlate clinically.
First MTP abrupt severe pain	Gout	Classic podagra but infection can still coexist.
DIP/PIP bony nodes	Hand OA	Heberden = DIP; Bouchard = PIP.
MCP/wrist inflammatory swelling	RA or CPPD/hemochromatosis mimic	Ordinary primary OA favors DIP/PIP/CMC more than MCP/wrist.
Widespread pain, fatigue, sleep/cognitive symptoms, normal labs	Fibromyalgia	Do not treat with steroids/immunosuppression.

Lateral hip pain lying on side	Greater trochanteric pain syndrome	Often gluteal tendinopathy rather than hip joint arthritis.
Pain with active/resisted motion but preserved passive ROM	Tendon/bursa/periarticular disease	Localize before ordering autoimmune panels.
Hot swollen joint in immunosuppressed patient	Septic arthritis until proven otherwise	Aspirate and culture; crystals do not exclude infection.

12. Medication Safety Crosswalk

The same medications recur across gout, CPPD, OA, and regional pain syndromes, but comorbidities decide safety. NSAIDs are often effective for crystal flares and OA pain but can worsen kidney function, blood pressure, heart failure, GI bleeding, and anticoagulation-related bleeding. Colchicine is useful for gout and CPPD but becomes dangerous with renal impairment and interacting drugs. Steroids rapidly reduce inflammation but cause hyperglycemia, mood changes, fluid retention, hypertension, osteoporosis, avascular necrosis, cataracts, adrenal suppression, and infection risk with repeated or chronic use. Local injections reduce systemic exposure but should not be injected through infected skin or into a joint/bursa where infection has not been excluded.

Medication	Useful for	Avoid/caution	High-yield adverse effects
NSAIDs	Acute gout/ CPPD, OA pain, bursitis/tendinopathy short courses.	CKD, GI bleed/ulcer, anticoagulation, heart failure, uncontrolled HTN, high CV risk.	AKI, edema, HTN, GI bleeding, CV events.
Colchicine	Acute gout early, flare prophylaxis, recurrent CPPD selected cases.	Advanced CKD/hepatic disease, strong CYP3A4/P-gp inhibitors, frailty.	Diarrhea, cytopenias, neuromyopathy, severe toxicity with interactions.
Glucocorticoids	Crystal flares when NSAID/colchicine unsafe; local OA/bursitis injections selected.	Untreated infection, poor diabetes control, frequent repeated injections, high osteoporosis risk.	Hyperglycemia, mood, insomnia, infection, osteoporosis, AVN, adrenal suppression.
Topical NSAIDs	Hand/knee OA, superficial regional pain.	Broken skin, NSAID allergy; systemic caution still in high-risk patients.	Local rash; lower systemic risk than oral NSAIDs.
Duloxetine	OA pain with centralized features; fibromyalgia.	Severe liver disease, substantial alcohol use, interacting serotonergic drugs, uncontrolled HTN.	Nausea, insomnia/somnolence, sweating, BP changes, withdrawal symptoms.
Pregabalin/gabapentinoids	Fibromyalgia phenotype with sleep/neuropathic features.	Sedation/falls, renal dose adjustment, misuse risk.	Dizziness, edema, cognitive slowing, weight gain.

13. Red Flags and Referral Triggers

Immediate or urgent evaluation is required for suspected septic arthritis, prosthetic joint infection, rapidly progressive joint destruction, acute compartment-like pain, acute neurologic deficit, cauda equina symptoms, vision loss, suspected fracture, systemic toxicity, or unexplained high inflammatory markers with focal pain. Rheumatology referral is appropriate for recurrent or atypical crystal arthritis, suspected inflammatory arthritis, unclear erosive disease, refractory gout needing advanced urate-lowering strategy, CPPD mimicking RA, or widespread pain with objective inflammatory features. Orthopedic referral is appropriate for advanced OA with severe disability, suspected tendon rupture, severe structural shoulder/hip/knee disease, or neurologic compression requiring procedural management.

Do not miss: septic arthritis hidden behind gout history; hip OA presenting as knee pain; CPPD presenting as wrist/MCP pseudo-RA; fibromyalgia coexisting with autoimmune disease; rotator cuff disease mislabeled as shoulder arthritis; trochanteric pain mislabeled as hip joint OA; carpal tunnel from inflammatory tenosynovitis; steroid complications from repeatedly treating noninflammatory pain as autoimmune flare.

14. Graph-RAG Concept Map

Node	Core relationships
Acute monoarthritis	Connected to septic arthritis, gout, CPPD, hemarthrosis, trauma; diagnostic edge: arthrocentesis.
Gout	Connected to hyperuricemia, CKD, diuretics, podagra, tophi, urate-lowering therapy, colchicine, NSAIDs, steroids.
CPPD	Connected to chondrocalcinosis, knee/wrist arthritis, hemochromatosis, hyperparathyroidism, hypomagnesemia, colchicine, OA mimic.
Osteoarthritis	Connected to mechanical pain, osteophytes, joint-space narrowing, weight loss, exercise, topical NSAIDs, arthroplasty.
Fibromyalgia	Connected to widespread pain, sleep disturbance, central sensitization, duloxetine/pregabalin, CBT, exercise, normal ESR/CRP.
Regional pain syndrome	Connected to bursitis, tendinopathy, entrapment neuropathy, preserved passive ROM, physical therapy, local injection.
Steroid exposure	Connected to hyperglycemia, infection risk, osteoporosis, AVN, adrenal suppression; medication safety edge.

15. Chapter Summary

Crystal arthritis, osteoarthritis, fibromyalgia, and regional pain syndromes are high-volume conditions that require disciplined localization. Acute crystal arthritis is inflammatory and can look infected; aspiration is the central diagnostic tool when uncertainty or risk exists. Gout is a urate-burden disease managed acutely with anti-inflammatory therapy and chronically with treat-to-target urate lowering when indicated. CPPD is an age-associated crystal disease with no crystal-dissolving therapy, so treatment targets flares, associated metabolic disease, and overlapping OA. OA is a whole-joint mechanical-structural disorder treated with exercise, weight and biomechanical interventions, topical/oral medications when safe, injections for selected flares, and arthroplasty for advanced disability. Fibromyalgia is centralized pain, not occult inflammation, and improves most reliably with education, graded activity, sleep and mood treatment, and selected neuromodulating medications. Regional pain syndromes are common mimics that localize to tendons, bursae, entheses, nerves, and periarticular tissues.

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